DATA ENGINEERING PROJECT

E-commerce Analytics Pipeline

From raw data to business insight with BigQuery, dbt and Power BI

Jaffle Shop dataset | January to April 2018 | November 2026

Business problem and objectives

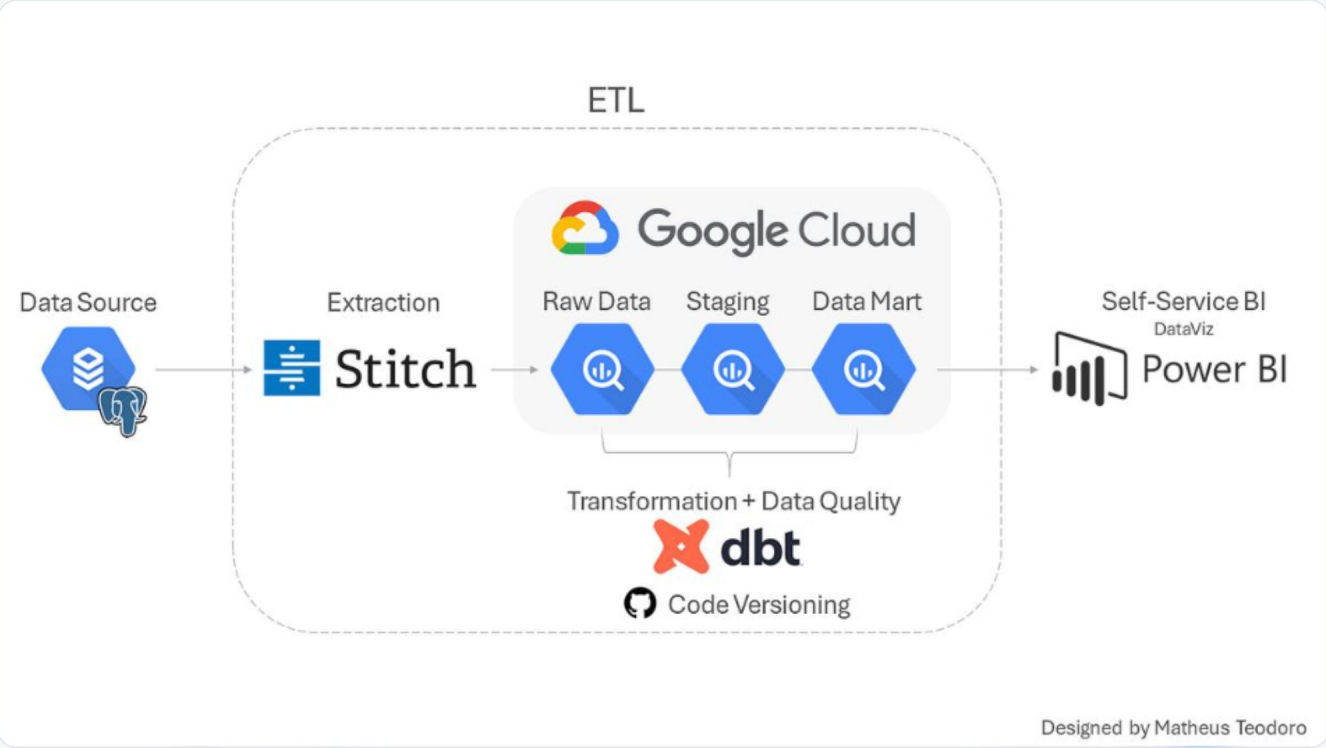
The problem

Raw operational tables are inconsistent, metrics such as a successful payment are not defined once, and nothing checks that the data is correct. Stakeholders cannot trust basic sales numbers.

Questions to answer

- How many orders and customers do we have?
- How much revenue comes from successful payments?
- Which customers and payment methods lead?
- How does revenue change over time?

Architecture



Raw tables loaded into BigQuery

Staging views clean and rename

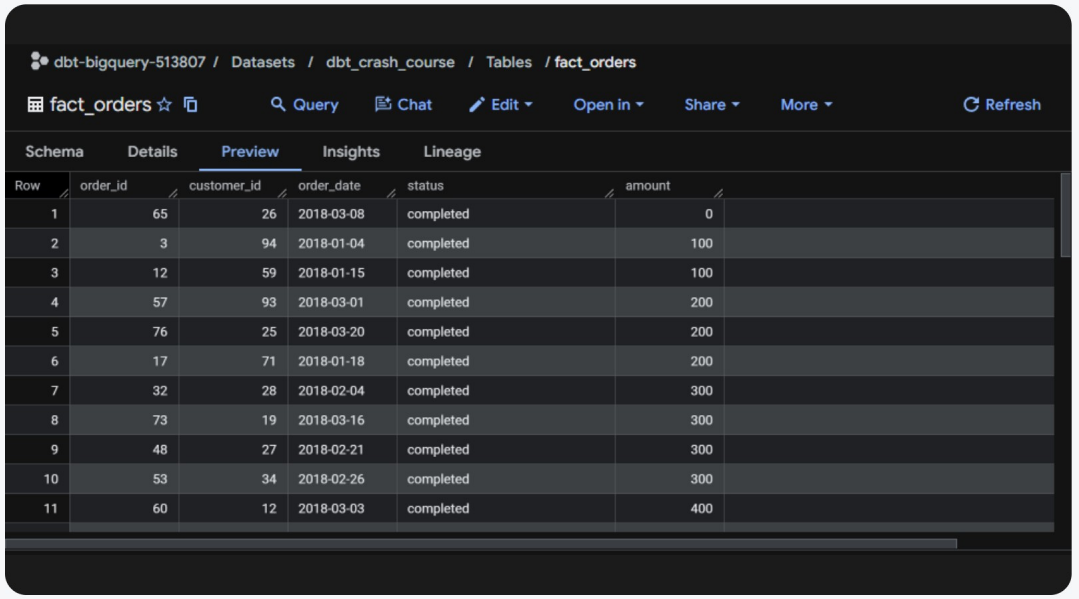
Marts tables serve Power BI

dbt transforms, tests and documents

Diagram adapted from Matheus Teodoro (Medium)

Data models

Model	Grain
dim_customers	One row per customer
fact_orders	One row per order
fact_payment	One row per payment



fact_orders in BigQuery

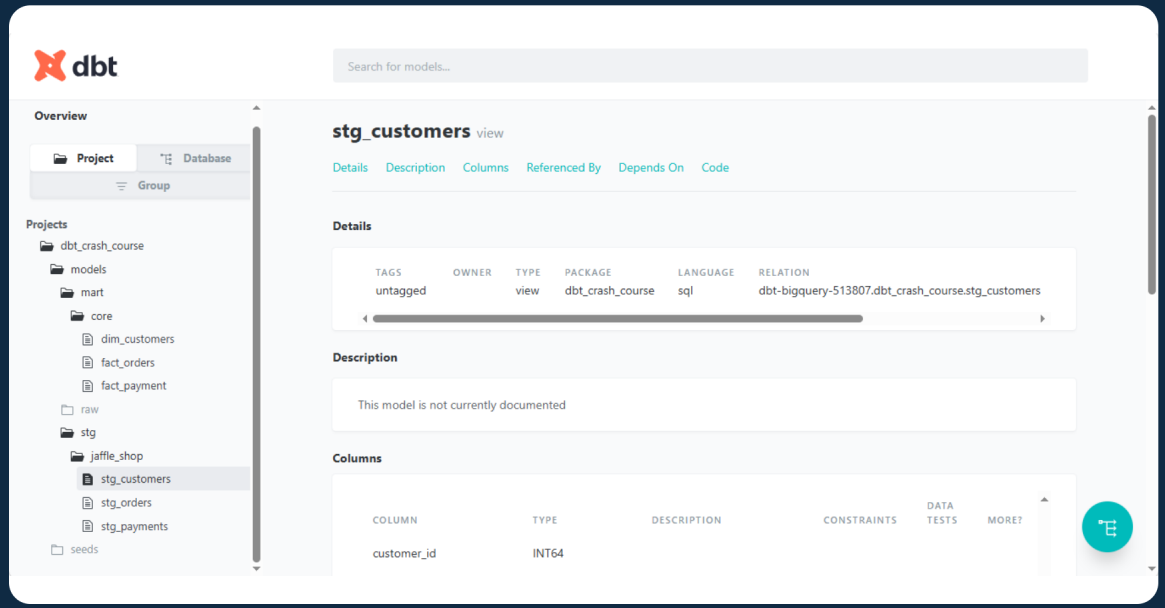
Staging models are views, mart models are tables

Data quality and documentation

Unique and not null
Every primary key is complete and distinct

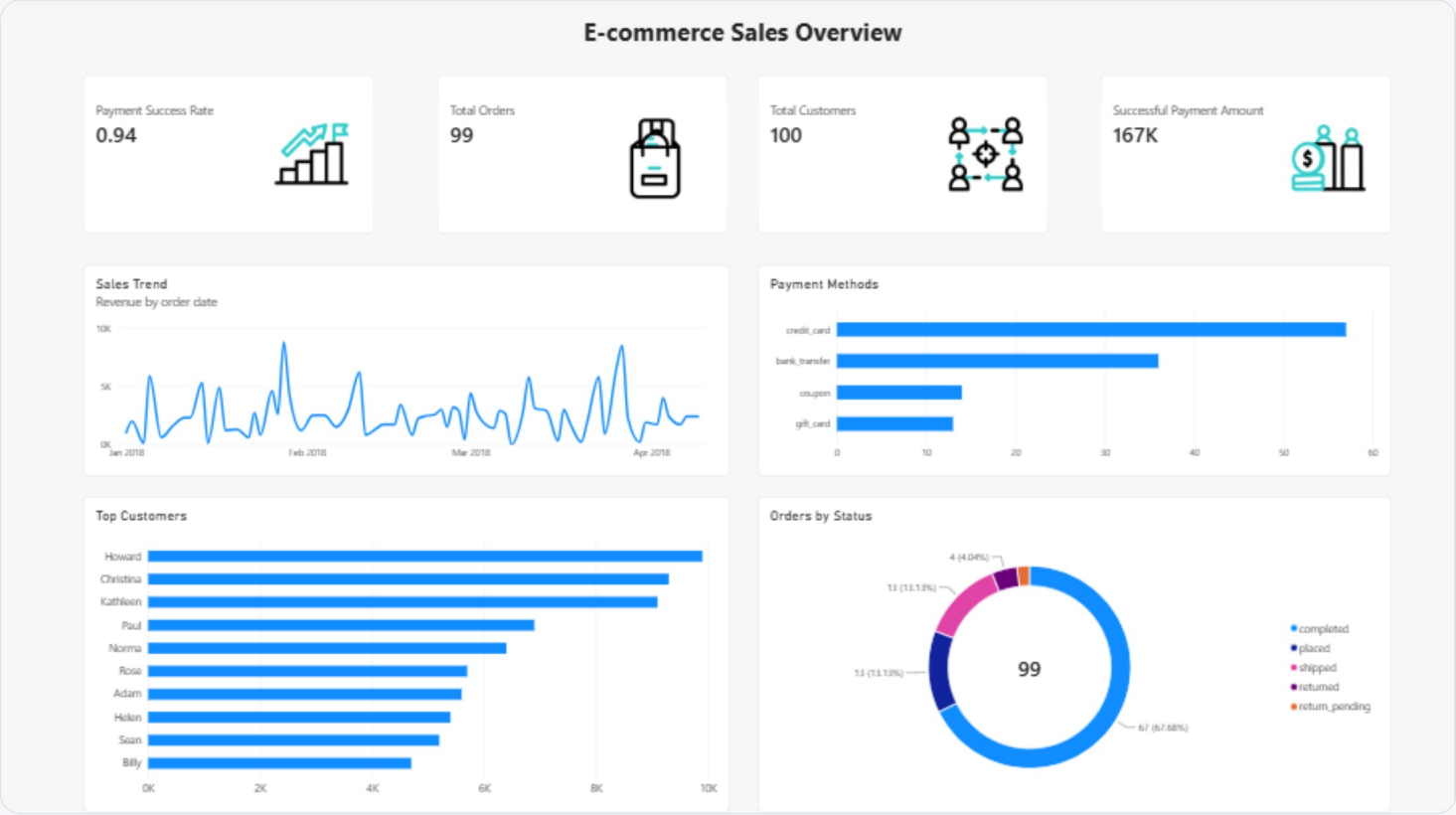
Relationships
Facts always point to a valid customer or order

dbt docs
Every model described, with a lineage graph



Tests run on every dbt build

Power BI dashboard



67 of 99 orders are completed

Credit card is the top payment method

Revenue is volatile, peaking near 10K

Built on the mart tables in BigQuery

Results and next steps

99	100	167K	0.94
Total orders	Total customers	Successful payments	Payment success rate

Next: scheduled dbt runs, CI on every pull request, incremental fact tables and date and product dimensions.